

Trigonometry & Polar Coordinates

ANSWER KEY



Law of Sines

- 1 In a triangle, $\angle A = 40^\circ$, $a = 10$, and $\angle B = 65^\circ$. Find b , rounded to the nearest hundredth.

By the Law of Sines, $\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{10 \sin 65^\circ}{\sin 40^\circ} \approx 14.10$.

- 2 In a triangle, $\angle A = 30^\circ$, $\angle B = 75^\circ$, and $a = 6$. Find $\angle C$ first, then explain how you would find side c .

$\angle C = 180^\circ - 30^\circ - 75^\circ = 75^\circ$. Then use $\frac{c}{\sin C} = \frac{a}{\sin A}$ to solve for c .

Law of Cosines

- 3 A triangle has $a = 10$, $b = 6$, and included angle $C = 40^\circ$. Find c , rounded to the nearest hundredth.

$c^2 = a^2 + b^2 - 2ab \cos C = 100 + 36 - 120 \cos 40^\circ \approx 44.07$, so $c \approx 6.64$.

- 4 Explain when you should use the Law of Cosines instead of the Law of Sines to solve a triangle.

The Law of Cosines is used when you know two sides and the included angle (SAS), or all three sides (SSS) — cases where the Law of Sines can't be set up directly because no side-angle pair is fully known.

Double-angle identities

- 5 If $\sin x = \frac{3}{5}$ and x is in Quadrant I (so $\cos x = \frac{4}{5}$), find $\sin 2x$ and $\cos 2x$.

$\sin 2x = 2 \sin x \cos x = 2 \left(\frac{3}{5}\right) \left(\frac{4}{5}\right) = \frac{24}{25}$. $\cos 2x = \cos^2 x - \sin^2 x = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$.

- 6 If $\cos x = -\frac{1}{2}$ and x is in Quadrant II (so $\sin x = \frac{\sqrt{3}}{2}$), find $\cos 2x$.

$\cos 2x = \cos^2 x - \sin^2 x = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$.

Polar to rectangular conversion

- 7 Convert the polar point $(8, 150^\circ)$ to rectangular coordinates.

$x = 8 \cos 150^\circ = 8 \left(-\frac{\sqrt{3}}{2}\right) = -4\sqrt{3}$, $y = 8 \sin 150^\circ = 8 \left(\frac{1}{2}\right) = 4$. So the point is $(-4\sqrt{3}, 4)$.

- 8 Convert the polar point $\left(6, \frac{\pi}{3}\right)$ to rectangular coordinates.

$x = 6 \cos \frac{\pi}{3} = 6 \left(\frac{1}{2}\right) = 3$, $y = 6 \sin \frac{\pi}{3} = 6 \left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$. So the point is $(3, 3\sqrt{3})$.

Rectangular to polar conversion

- 9 Convert the rectangular point $(-3, 3\sqrt{3})$ to polar coordinates (r, θ) with $r > 0$ and $0^\circ \leq \theta < 360^\circ$.

$$r = \sqrt{(-3)^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = 6.$$

Since the point is in Quadrant II, $\theta = 120^\circ$ (from $\tan \theta = \frac{3\sqrt{3}}{-3} = -\sqrt{3}$, adjusted for the correct quadrant).